

```

1906         for index in range(1, info["count"] + 1)
1907     }
1908     relay_unit_ready = {"R01": 0.0, "R02": 0.0}
1909     pack_ready = {"R-PACK-{index:02d}": 0.0 for index in range(1, data["relay_inventory"]["packs"] + 1)}
1910     q3_trips: list[dict[str, Any]] = []
1911     relay_missions: list[dict[str, Any]] = []
1912     communication_rows: list[dict[str, Any]] = []
1913     proof_summaries: list[dict[str, Any]] = []
1914     relay_type = data["relay_type"]
1915
1916     for wave_name, trip_indices, point_names in wave_definitions:
1917         selected = [q2_trips[i] for i in trip_indices]
1918         assignments = []
1919         reserved_packs: set[str] = set()
1920         for point_index, point_name in enumerate(point_names):
1921             unit_id = f"R{point_index + 1:02d}"
1922             eligible_packs = [p for p in pack_ready if p not in reserved_packs]
1923             pack_id = min(
1924                 eligible_packs,
1925                 key=lambda p: (max(pack_ready[p], relay_unit_ready[unit_id]), p),
1926             )
1927             reserved_packs.add(pack_id)
1928             resource_ready_s = max(relay_unit_ready[unit_id], pack_ready[pack_id])
1929             point = points[point_name]
1930             preliminary = relay_flight_metrics(dem, depot, relay_type, *point, 0.0)
1931             assignments.append(
1932                 {"point_name": point_name, "point": point, "unit_id": unit_id, "pack_id": pack_id,
1933                  "resource_ready_s": resource_ready_s, "preliminary": preliminary}
1934             )
1935             common_start = max(
1936                 [unit_ready[x["unit_id"]] for x in selected]
1937                 + [battery_ready[x["battery_id"]] for x in selected]
1938                 + [0.0]
1939             )
1940             if assignments:
1941                 min_ground_prep = min(
1942                     types[x["type_id"]].prep_s + len(x["box_ids"]) * types[x["type_id"]].load_per_box_s
1943                     for x in selected
1944                 )
1945                 common_start = max(
1946                     common_start,
1947                     max(
1948                         x["resource_ready_s"] + x["preliminary"]["link_ready_offset_s"]
1949                         - min_ground_prep
1950                         for x in assignments
1951                     ),
1952                 )
1953             for assignment in assignments:
1954                 assignment["start_s"] = (
1955                     common_start + min_ground_prep
1956                     - assignment["preliminary"]["link_ready_offset_s"]
1957                 )
1958                 assignment["link_ready_s"] = (
1959                     assignment["start_s"]
1960                     + assignment["preliminary"]["link_ready_offset_s"]
1961                 )
1962
1963     wave_trips: list[dict[str, Any]] = []
1964     for original in selected:
1965         g = types[original["type_id"]]
1966         timeline = route_timeline(
1967             dem, depot, nodes, g, boxes, _trip_stops_from_record(original, boxes), common_start
1968         )
1969         if timeline is None:
1970             raise RuntimeError(f"Q3 shifted route became infeasible: {original['trip_id']}")
1971         charge_duration = charge_time_s(
1972             timeline["return_soc"], data["battery_inventory"][original["type_id"]]["full_charge_s"]
1973         )
1974         shifted = {
1975             **original,

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